

Building a Censoring Pipeline for Movie Trailers and Movie Clips

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Abstract

This paper describes a pipeline built in Python that takes in videos of varying lengths and applies a filtering system. Whisper AI is used to transcribe the audio and flags chosen profanity. Replacement words are reinserted into the audio using edge-tts and then the audio and video are merged together to create one single censored mp4 file. While it does have base functionality, improvements such as better volume and timestamp matching could be added to the pipeline for an improved and efficient filtering system.

1. Introduction and Literature Review

M.A.S.H. (1970) was the first Hollywood film to have the f-word in it (Malas 2023). Since then, many movies have used that word and other curse words. At the same time, television was becoming big, which led to broadcast tv versions of popular movies that edited out excessive profanity, sexual content, and violence. In the modern age, cable tv is not as popular as it once was, with streaming services being the preferred way for audiences to view movies and tv shows at home. Filtering services like VidAngel go through the movies and episodes on streaming services and mark inappropriate language and other offensive content so that users can choose to filter it out or not. The issue with their language filtering system is that it simply mutes the word, instead of replacing it with a substitute word, which is what broadcast tv edits do. In extreme cases, audio surrounding the word is muted as well, leading to the muting of entire scenes. This can take the viewer out of the experience and makes the audio sound choppy and unprofessional.

Modern approaches to filtering include using deep learning models for profanity detection. It is extremely time intensive for humans to go through and mark every instance of

profanity in a piece of content, and human error is still possible. In their paper, Wazir et. al created a convolutional neural network pipeline that can be used for automated profanity detection. They created a large dataset of profanities for others to use that can make this process more efficient for everyone. Their study included YouTube clips and feature-length movies with varying levels of profanity (Wazir et al. 2022).

My filtering system is meant to be similar to a broadcast tv edit of a movie. In addition, it is similar to Wazir et al.'s framework, albeit at a much smaller scale. For the purposes of this smaller scale project, I have filtered two movie trailers and one full-length movie clip that have varying levels of profanity to test how well my system can perform when a video is longer and when a video has high amounts of profanity. While I do not have specific research questions, my project sets out to solve the issue of making a semi-automated censoring pipeline for the modern age of streaming. While my system is relatively barebones, it has potential to improve the viewing and listening experience for audiences. As someone who loves being immersed in the viewing experience, I find this to be an important issue to resolve.

2. Methods

I used the yt-dlp tool to scrape the trailers for *Lanterns* and *Wild Horse Nine* and a movie clip from *One Battle After Another*. I chose these videos because they all have either a variety of profanity or high amounts of it, so they would be good ways to show if my system works well or not. My system used OpenAI's Whisper software for the transcriptions and edge-tts, a python package for Microsoft's TTS system, for the audio replacement. I created a python dictionary of bad words that I wanted to be replaced with clean words, which can be seen in my code. I used both the "base" and "medium" models of Whisper to improve transcription quality if the system was not properly picking up certain words. For each video, I calculated the precision, recall, and

F1 score. In addition, the total number of bad words were calculated. The number of words detected and matched were then calculated, along with any missed words or false positives. These metrics were all calculated and tallied based off the Whisper transcript created in my script.

My system outputted various versions of the video and audio in mp4 and wav formats. The user can watch the original video with the original audio, the fully censored video or audio, or they can choose which track they want to listen to with the stereo option. The stereo option must be watched with headphones on. The left earphone has the uncensored audio, while the right earphone has the censored audio.

This entire system runs in a Python script that was partially written by me, Claude, and ChatGPT. The full pipeline takes in the originally scraped video and audio files, then finds all instances of the bad words, takes them out and replaces them with the clean words, and then outputs the original audio and video, the clean audio and video, and the stereo audio and video for the user to download to their hard drive for viewing on their personal device.

3. Results

For the *Wild Horse Nine* trailer, there were a total of 8 words flagged in the Whisper transcript and each word was matched correctly, leading to perfect precision, recall, and F1 scores. The *Lanterns* trailer had 3 words and similarly had perfect evaluation metrics. The movie clip from *One Battle After Another* had 58 instances of bad language, which makes sense considering this clip was four times longer than the trailers, and a full unedited movie, instead of a shorter trailer that may still be edited to a degree. Again, the metrics came out perfect. This provides strong evidence that my model is able to produce consistent and reliable filtering.

However, it is important to note that these metrics were all based off of the Whisper transcript, which is not a perfect metric. In the *Wild Horse Nine* trailer, there is a moment where the word “Czechoslo-f*cking-vakia” appears. Since the f-word is in the middle of another word, it is not flagged and therefore was not matched properly. Eight out of nine words is still very good, but the system still needs some fine-tuning. One potential fix would be to manually enter that new compound word as something to be replaced, but the Whisper “medium” model transcribes that compound word as “Czechoslovakia”, so that may not fix the issue.

In some runs, I implemented a fuzzy matching approach, so that the model could flag any words that are similar to the curse words it was looking for. While that sounds like it would work well in practice, it actually over implemented the filtering algorithm. Any time the word “shot” was said, it was replaced with “crap”, which also breaks immersion. That was the issue I was trying to work away from. For the final runs, I decided to take away that functionality, as it appeared to be hurting the model’s performance more than helping it.

4. System Evaluation

While my model does show consistently strong results, it is still a very basic system. It is clearly good at finding and replacing words, but sometimes there are volume or timing mismatches. For example, the replacement words are sometimes louder than the original audio, making the replacement sound stilted and out of place, even if it was filtered correctly. With regards to the timing mismatches, there are sometimes instances where the beginnings of the curse word can be heard before the replacement audio begins. It does not happen very often in the system, but that is an issue. If someone were to use this system hoping for perfect filtering, they would not be completely happy with the results. However, given that this is an early attempt that could be described as a prototype, I think it works well. I had someone outside of the class

view the censored trailer for *Wild Horse Nine*, and they were impressed with how well it worked. They did say that the model missed some words, but that when the filtering was working correctly, it sounded natural.

5. Conclusion and Future Directions

This small-scale project has proved that there is a better way to create filtering systems that enables viewers to enjoy filtering if they choose to do so, without experiencing the frustrations that come from systems where words or whole scenes are completely muted. While systems like VidAngel are very robust, they are also very greedy with their filtering systems. My system works well for what it is, but it could still be improved. If I continue to work on this system in the future, the first thing to fix would be having better volume control and timestamp matching. In addition, I would add in AI voice cloning to more closely mimic the voice of the original speaker in the audio. Regarding my profanity “library”, I focused only on single words. There are profanity phrases as well that would need to be handled differently in a full system. This system was currently only tested on short movie trailers and clips, so it would be interesting to see how it performs on a full-length movie or tv episode.

In addition, I might make a machine learning model that decides if a word should be replaced or simply muted. For example, it is very easy to filter the standard library of curse words, but for instances of profanity that include sexual references, it may be easier to simply mute the scene or cut it out entirely. This could be done using contextual clues, which would also be a good idea for curse words that have different meanings in different contexts, like “b*tch.” Some research has begun to look at using machine learning to censor social media content, which could be another way that this system is used in the future. One research team used Whisper transcripts to censor audio and merge it back together with the video, similar to my

system. However, they did so for social media, instead of movies (Kumar, Bansal & Subburaj 2025). One could argue that filtering is an even more important issue on social media sites, since everything is live and completely unfiltered. Movies and tv, on the other hand, are not uncensored in the same way.

Overall, I was surprised by how well this small system was able to accurately find the bad words and replace them with their clean counterparts. It is not perfect, but it shows that tools like Whisper's transcription and TTS models can be used well together to create filtering pipelines.

Works Cited

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